

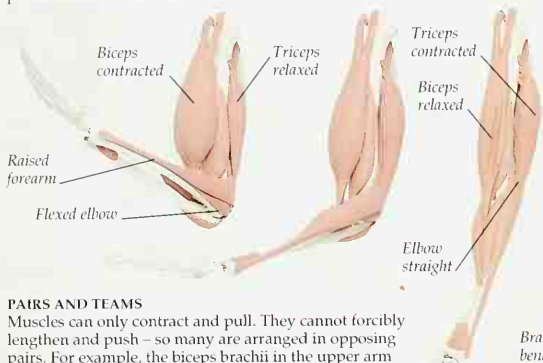


The moving body

BY THE LATE 18TH CENTURY, virtually all of the body's muscles had been identified and named. They are invariably in mirror-image pairs, emphasizing the symmetrical nature of the skeleton, as opposed to the inner organs such as the intestines. The biggest muscle is the gluteus maximus (p. 21) in the buttock. The smallest muscle is the stapedius, attached to the smallest bone, the stapes or stirrup (p. 55) inside the ear. Like a short piece of cotton thread, it pulls on the stapes to dampen the vibrations from too-loud sounds, preventing them from causing damage to the delicate inner ear. Once general muscle anatomy was complete, microscopists and physiologists then discovered exactly how muscles work. Jan Swammerdam (1637-1680) showed that muscles alter in shape, not in size, during contraction. Luigi Galvani (1737-1798) (p. 58) noted the link between muscles and electricity. Subsequent research showed that muscles contract on stimulation by the tiny electrical impulses of nerve signals.

THE THREE "S" WORDS

Muscle fitness can be assessed by three S-words: strength, stamina, and suppleness. Some exercises develop only one factor, but moving activities like dancing and swimming promote all three factors.

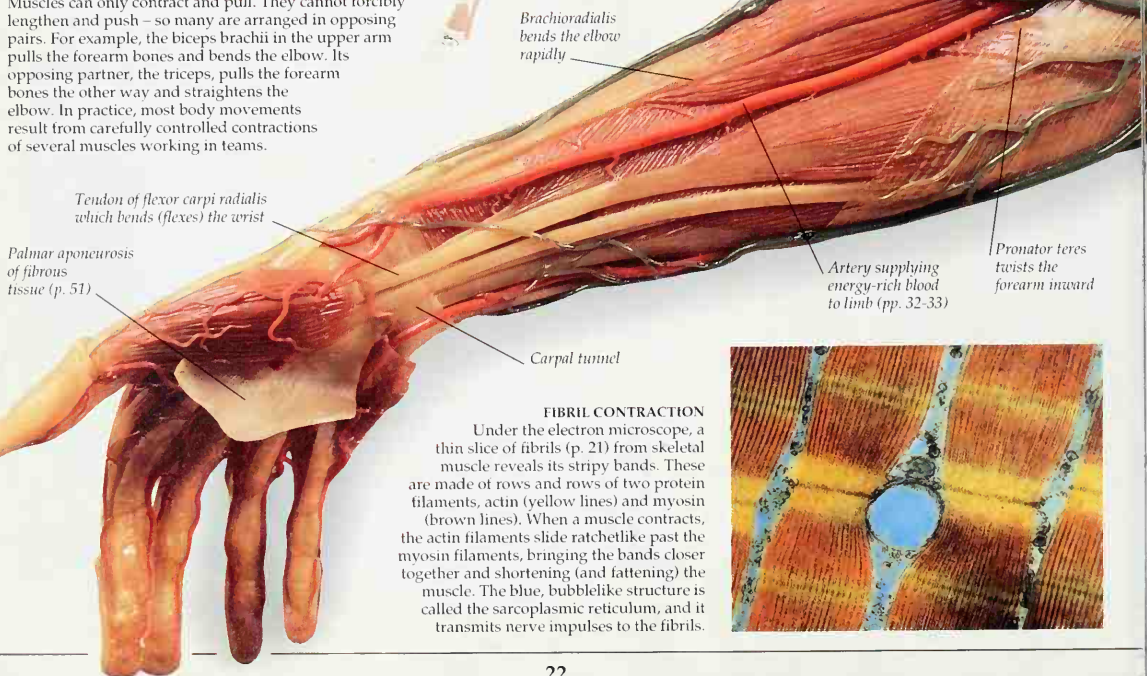


PAIRS AND TEAMS

Muscles can only contract and pull. They cannot forcibly lengthen and push – so many are arranged in opposing pairs. For example, the biceps brachii in the upper arm pulls the forearm bones and bends the elbow. Its opposing partner, the triceps, pulls the forearm bones the other way and straightens the elbow. In practice, most body movements result from carefully controlled contractions of several muscles working in teams.

MOVING THE ARM AND HAND

Muscles in the upper chest and back move the shoulder; those in the upper arm move the elbow; those in the forearm move the wrist and fingers. The leg has a very similar overall design. Traditionally, anatomists call the relatively "fixed" end of the muscle, nearest the center of the body, the origin of the muscle, and the farther end, which "moves," the insertion.



FIBRIL CONTRACTION

Under the electron microscope, a thin slice of fibrils (p. 21) from skeletal muscle reveals its stripy bands. These are made of rows and rows of two protein filaments, actin (yellow lines) and myosin (brown lines). When a muscle contracts, the actin filaments slide ratchetlike past the myosin filaments, bringing the bands closer together and shortening (and fattening) the muscle. The blue, bubblelike structure is called the sarcoplasmic reticulum, and it transmits nerve impulses to the fibrils.

